

Enterprise Regulatory RAG — Technical Solution

15 PDFs • 361 pages • 2,534 chunks • evaluated retrieval + grounded generation

Routed Hit@5 92.3%

Abstention accuracy 100%

OFFLINE / INDEXING

15 PDFs

regulatory + clinical

HASH

SHA-256 incremental

LOAD + CLEAN

parse • normalize

CHUNK

structure-aware 500/50

EMBED

text-embedding-3-small

QDRANT

vectors + metadata

NEW → index • UPDATED → replace • UNCHANGED → skip • REMOVED → delete

ONLINE / QUERY

USER / SESSION

history + filters

REWRITE*

follow-up only

ACL / FILTER

document / tenant

RETRIEVAL ROUTER

Semantic → Dense
Exact CFR → BM25 + local context

TOP-K

trusted evidence

GROUNDING LLM

GPT-4.1-mini
evidence-only • abstain

"What about alternatives?" → standalone retrieval query

ANSWER + [1][2]

WHY THIS DESIGN?

500/50 selected empirically
Dense for semantic queries
BM25 fixes exact CFR lookup
RRF tested; routing won overall

PRODUCTION READINESS

Incremental SHA-256 lifecycle
Deterministic chunk IDs
Metadata / ACL filtering
FastAPI + sessions + cached retriever

QUALITY & SAFETY

Evidence-only generation
Backend-owned provenance
Citation validity 100%
Abstention accuracy 100%

OBSERVABILITY → OPTIMIZATION

Rewrite
tracked

Retrieval
route + latency

Generation
latency

Tokens
tracked

Citations
validated

Cold setup dominated by BM25 build; cached retriever makes warm setup ~0 ms

Retrieval Evaluation

RETRIEVAL EVALUATION — BASELINE

500 / 50 delivered the strongest chunking baseline

Chunk size / overlap experiment performed before expanding to the final 15-document corpus

CONFIGURATION	HIT@1	HIT@3	HIT@5	MRR
500 / 50	80%	100%	100%	0.9000
800 / 100	80%	100%	100%	0.8889
1200 / 150	73.33%	93.33%	93.33%	0.8333

What the experiment shows

500-character chunks with 50-character overlap preserved enough context while remaining focused. This baseline was retained when the corpus expanded.

Selected baseline: 500 / 50

Query-aware routing produced the strongest overall retrieval

Expanded benchmark with semantic, exact regulatory, and cross-document questions

RETRIEVER	HIT@1	HIT@3	HIT@5	MRR
Dense	64.10%	79.49%	82.05%	0.7115
BM25	64.10%	82.05%	82.05%	0.7313
Hybrid RRF	61.54%	79.49%	87.18%	0.7201
Routed	71.79%	89.74%	92.31%	0.7970

The practical decision

Dense remains the semantic default. Exact CFR section/subsection queries route to BM25, then expand local context to reconnect footnotes with answer-bearing prose.

Semantic → Dense • Exact CFR → BM25 + context expansion

Grounding stayed strong on the harder 15-document benchmark

24 questions: 18 answerable + 6 unanswerable • answer quality separated from faithfulness and citation quality

DIMENSION	PRIMARY	RATE / CHECK	SCORE	OUTCOME
Answer quality	Terms 94.44%	Citations 94.44%	2.72 / 3	Strong
Faithfulness	Faithful answers	94.44%	2.83 / 3	Grounded
Citations	Complete 96.43%	Valid 100%	2.62 / 3	86.39% fully correct
Abstention	False abstain 5.56%	Accuracy 100%	—	Safe refusal

What the benchmark shows

The expanded benchmark exposed one false abstention and a few citation-selection gaps. Citation IDs remained 100% valid, while semantic citation correctness was intentionally stricter than simple ID validation.

Credible evaluation > tuning every metric to 100%